

Use the Taylor expansion to find the linear approximation $L(x)$ to $F(x)$ near x_0 , where

$$F : \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \quad \begin{pmatrix} u \\ v \\ w \end{pmatrix} \longmapsto \begin{pmatrix} u + e^{u+w} \\ v + w^2 \\ uv + w \end{pmatrix}, \quad x_0 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$